

MECHATRONICS AND IOT

ASSIGNMENT REPORT - 6

TITLE:

Design and simulate a Smart Refrigerator Monitoring System using DHT11 Sensor. The system monitors internal temperature and humidity conditions to improve food preservation and energy efficiency.

DONE BY ,

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Project Overview:

The Smart Refrigerator Monitoring System is an IoT-based system designed to continuously monitor the internal temperature and humidity of a refrigerator to maintain suitable conditions for food preservation. The system uses a DHT11 sensor to measure the temperature and humidity inside the refrigerator, while an ESP8266 NodeMCU processes the collected sensor data.

The measured parameters are displayed on an OLED display, allowing users to monitor the refrigerator conditions in real time. Based on predefined threshold values, the system can identify Normal and Abnormal conditions. An LED and buzzer can provide an alert when the temperature or humidity exceeds the desired range.

The system can also use the ESP8266's Wi-Fi connectivity to transmit the monitored data to an IoT/cloud platform for remote monitoring and analysis. The proposed system provides a low-cost, real-time and automated solution for improving food preservation, reducing spoilage and supporting efficient refrigerator operation.

Problem Statement:

Maintaining proper temperature and humidity conditions inside a refrigerator is essential for preserving food quality and preventing spoilage. Conventional refrigerators may not provide continuous monitoring or immediate alerts when the internal conditions move outside the desired range.

Temperature fluctuations, excessive humidity or prolonged exposure to unsuitable conditions can reduce food shelf life and increase food wastage. Manual checking of refrigerator conditions is also inconvenient and may result in delayed detection of abnormal conditions.

Therefore, there is a need for a low-cost, real-time refrigerator monitoring system that can continuously measure temperature and humidity, identify abnormal conditions and provide timely alerts to improve food preservation, safety and energy-efficient refrigerator operation.

Proposed Solution:

The proposed system uses a DHT11 sensor and ESP8266 NodeMCU to continuously monitor the temperature and humidity inside the refrigerator. The DHT11 sensor

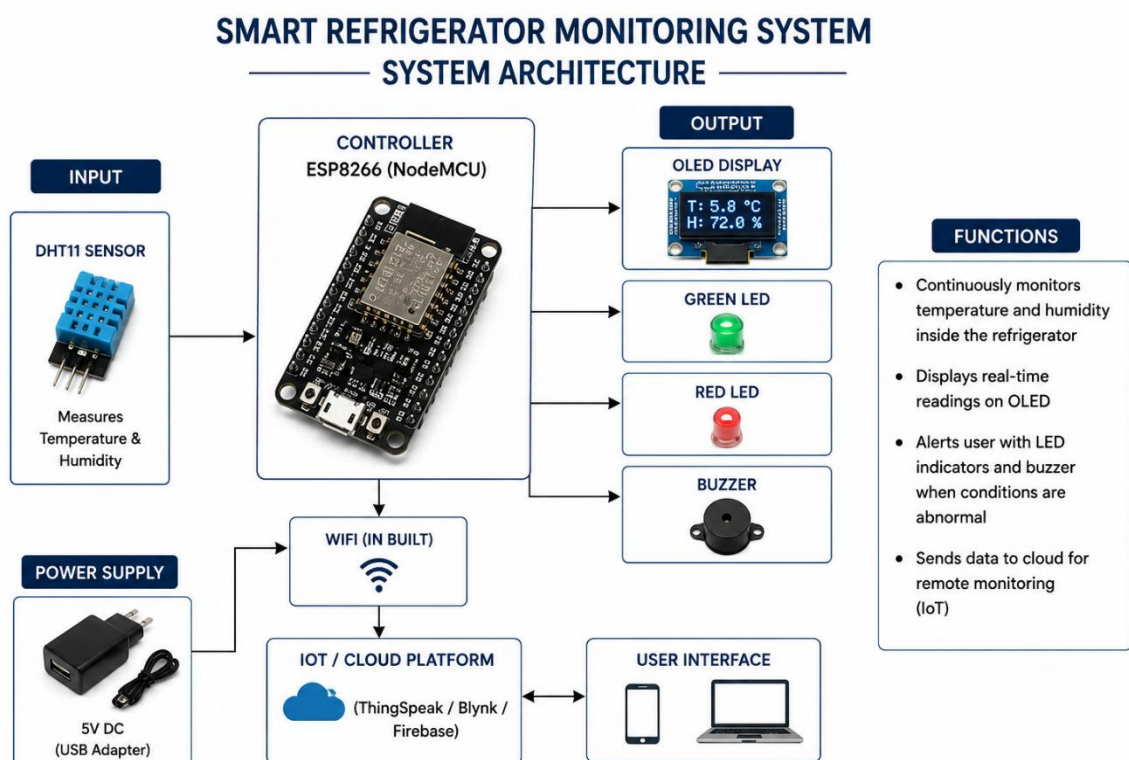
measures the internal environmental conditions and sends the readings to the ESP8266 for processing.

The ESP8266 compares the measured temperature and humidity with predefined threshold values to determine whether the refrigerator is operating under Normal or Abnormal conditions. The current readings and status are displayed on an OLED display for real-time local monitoring.

When the temperature or humidity exceeds the predefined safe range, a red LED and buzzer are activated to alert the user. Under normal conditions, a green LED indicates that the refrigerator environment is suitable for food preservation.

The ESP8266 can also use its built-in Wi-Fi connectivity to transmit the sensor data to an IoT/cloud platform for remote monitoring and data analysis. The proposed system provides a low-cost, real-time and automated solution that helps improve food preservation, reduce spoilage and monitor refrigerator performance efficiently.

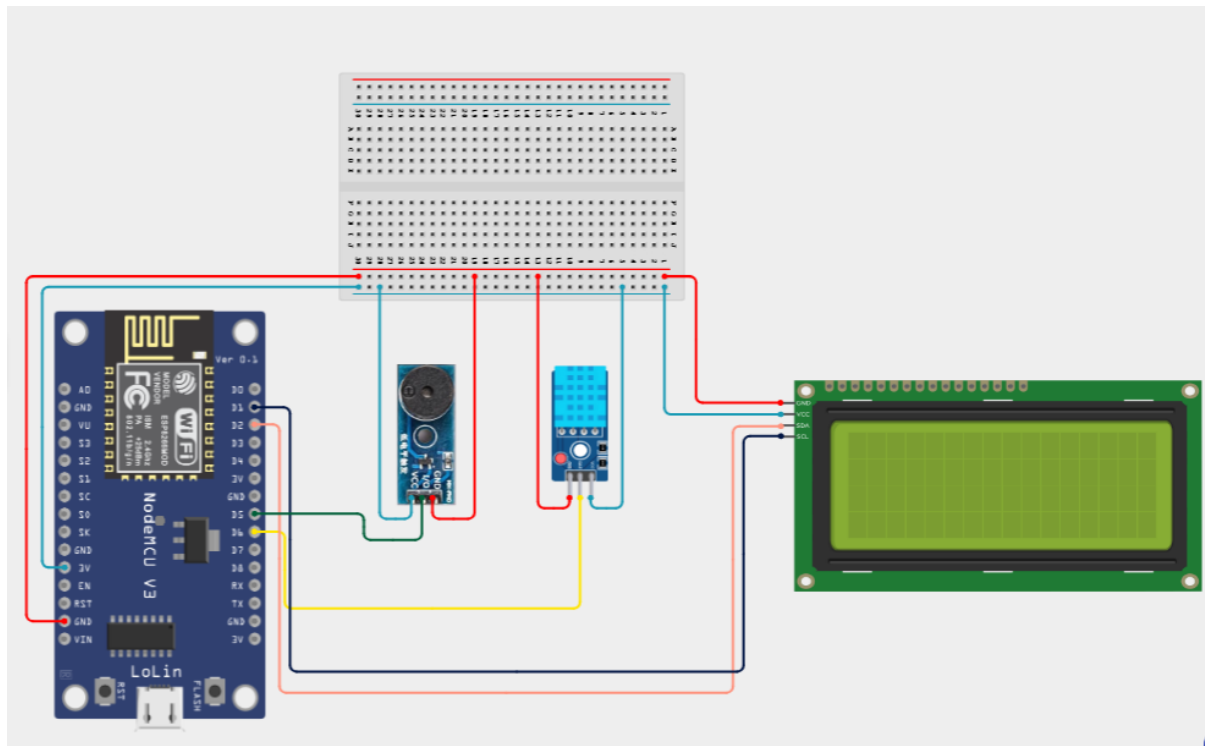
System Architecture:



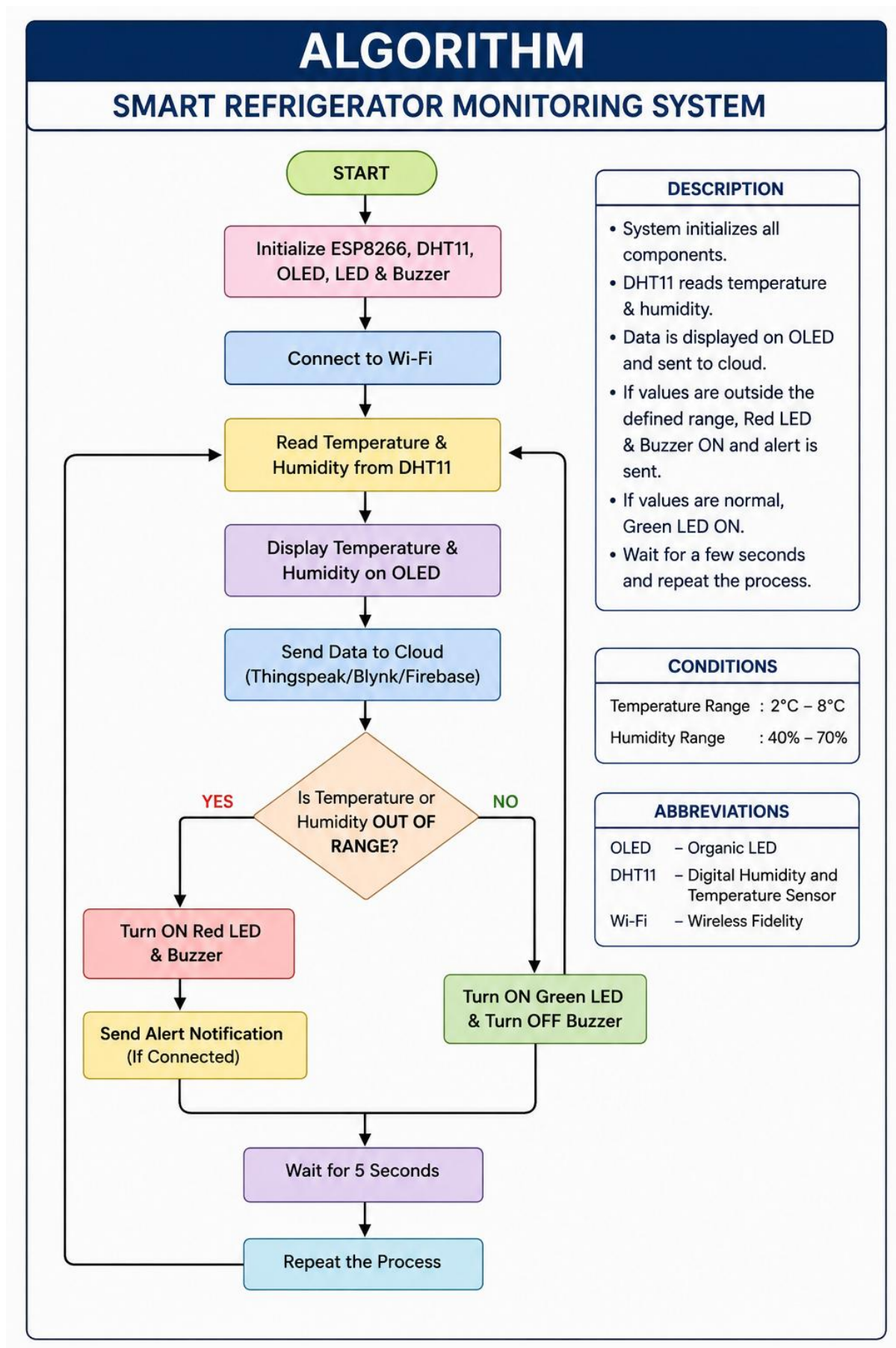
Circuit Table :

Component	Pin/Terminal	ESP8266 Connection	Purpose
DHT11 Sensor	VCC	3.3V	Power supply
DH T11 Sensor	DATA	D4 (GPIO 2)	Temperature & humidity data
LCD Display	VCC	3.3V	Safe indication
ESP8266	VIN/USB	5V USB	Hazardous indication
Buzzer	Positive (+)	GPIO 14	Alert

Circuit Design:



Algorithm:



Coding:

```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <DHT.h>

#include <ESP8266WiFi.h>

#include <ThingSpeak.h>

// =====

// WIFI SETTINGS

// =====

const char* ssid = "Xiaomi 11i";
const char* password = "12345678";
WiFiClient client;

// =====

// THINGSPEAK SETTINGS

// =====

unsigned long channelID = 3460409;
const char* writeAPIKey = "O8JFT9FMYPONB4PP";

// =====

// DHT11 SETTINGS

// =====

#define DHT_PIN 02
#define DHT_TYPE DHT11
DHT dht(DHT_PIN, DHT_TYPE);

// =====

// LCD SETTINGS

// =====

LiquidCrystal_I2C lcd(0x27, 16, 2);

// =====

// BUZZER SETTINGS
```

```
// =====  
#define BUZZER_PIN 14 // D5 = GPIO14  
// =====  
// SETUP  
// =====  
void setup()  
{  
  Serial.begin(115200);  
  // =====  
  // START BUZZER  
  // =====  
  pinMode(BUZZER_PIN, OUTPUT);  
  digitalWrite(BUZZER_PIN, LOW);  
  // =====  
  // START DHT11  
  // =====  
  dht.begin();  
  // =====  
  // START I2C  
  // SDA = GPIO4 (D2)  
  // SCL = GPIO5 (D1)  
  // =====  
  Wire.begin(4, 5);  
  // =====  
  // START LCD  
  // =====  
  lcd.init();  
  lcd.backlight();  
  lcd.clear();  
  lcd.setCursor(2, 0);
```

```
    lcd.print("HVAC MONITOR");
    lcd.setCursor(1, 1);
    lcd.print("Connecting...");
    // =====
    // CONNECT TO WIFI
    // =====
    WiFi.begin(ssid, password);
    Serial.println();
    Serial.print("Connecting to WiFi");
    while (WiFi.status() != WL_CONNECTED)
    {
        delay(500);
        Serial.print(".");
    }
    Serial.println();
    Serial.println("WiFi Connected!");
    Serial.print("IP Address: ");
    Serial.println(WiFi.localIP());
    // =====
    // CONNECT TO THINGSPEAK
    // =====
    ThingSpeak.begin(client);
    Serial.println("ThingSpeak Connected!");
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("WiFi Connected");
    lcd.setCursor(0, 1);
    lcd.print("Cloud Ready");
    delay(2000);
}
```

```

// =====
// LOOP
// =====

void loop()
{
    // =====
    // READ DHT11
    // =====

    float temperature = dht.readTemperature();
    float humidity = dht.readHumidity();

    // =====
    // CHECK SENSOR
    // =====

    if (isnan(temperature) || isnan(humidity))
    {
        Serial.println("DHT11 reading failed!");
        // Buzzer OFF during sensor error
        digitalWrite(BUZZER_PIN, LOW);
        lcd.clear();
        lcd.setCursor(2, 0);
        lcd.print("SENSOR ERROR");
        lcd.setCursor(0, 1);
        lcd.print("Check DHT11");
        delay(2000);
        return;
    }
    // =====
    // HVAC STATUS
    // =====

    int statusValue;

```

```
if (temperature > 30)
{
    statusValue = 2;
}
else if (temperature < 20)
{
    statusValue = 0;
}
else
{
    statusValue = 1;
}

// =====
// BUZZER CONTROL
// =====

if (temperature > 30 || temperature < 20)
{
    digitalWrite(BUZZER_PIN, HIGH);
}
else
{
    digitalWrite(BUZZER_PIN, LOW);
}

// =====
// SERIAL MONITOR
// =====

Serial.print("Temperature: ");
Serial.print(temperature);
Serial.println(" C");
Serial.print("Humidity: ");
```

```
Serial.print(humidity);
Serial.println(" %");
Serial.print("HVAC Status: ");
if (statusValue == 2)
{
    Serial.println("HIGH");
}
else if (statusValue == 0)
{
    Serial.println("LOW");
}
else
{
    Serial.println("NORMAL");
}
// =====
// LCD DISPLAY
// =====
lcd.clear();
// Line 1
lcd.setCursor(0, 0);
lcd.print("T:");
lcd.print(temperature, 1);
lcd.print((char)223);
lcd.print("C");
lcd.setCursor(9, 0);
lcd.print("H:");
lcd.print(humidity, 1);
lcd.print("%");
// Line 2
```

```
lcd.setCursor(0, 1);
if (temperature > 30)
{
    lcd.print("Status: HIGH");
}
else if (temperature < 20)
{
    lcd.print("Status: LOW");
}
else
{
    lcd.print("Status: NORMAL");
}

// =====
// SEND TO THINGSPEAK
// =====

ThingSpeak.setField(1, temperature);
ThingSpeak.setField(2, humidity);
ThingSpeak.setField(3, statusValue);
int response = ThingSpeak.writeFields(
    channelID,
    writeAPIKey
);
if (response == 200)
{
    Serial.println("Cloud upload successful!");
}
else
{
    Serial.print("Cloud upload failed. Error: ");
```

```

        Serial.println(response);
    }
    Serial.println("-----");
    // =====
    // WAIT
    // =====
    delay(20000);
}

```

System Working:

- The Smart Refrigerator Monitoring System uses a DHT11 sensor and ESP8266 NodeMCU to continuously monitor the temperature and humidity inside the refrigerator. The DHT11 sensor collects the environmental conditions and sends the readings to the ESP8266 for processing.
- The ESP8266 compares the measured values with predefined temperature and humidity limits. The current readings are displayed on the OLED display in real time.
- When the temperature or humidity remains within the desired range, the green LED indicates normal conditions. If either parameter exceeds the predefined limit, the red LED and buzzer are activated to alert the user about abnormal refrigerator conditions.
- The ESP8266 can also use its Wi-Fi connectivity to transmit the collected data to an IoT/cloud platform for remote monitoring and data analysis. The system continuously repeats the sensing and monitoring process to help maintain suitable conditions for food preservation and efficient refrigerator operation.

Bill Of Material:

S. no	Component	Specification	Quantity	Cost
1	ESP8266	Node MCU ESP8266	1	Rs. 450
2	DHT11 Sensor	Temperature & Humidity Sensor	1	Rs. 70
3	Buzzer	5V Active Buzzer	1	Rs. 50
4	Bread Board	Standard 830-point	1	Rs. 100

5	Jumper Wire	Male-Male / Male-Female	1 set	₹50
		Total		₹720

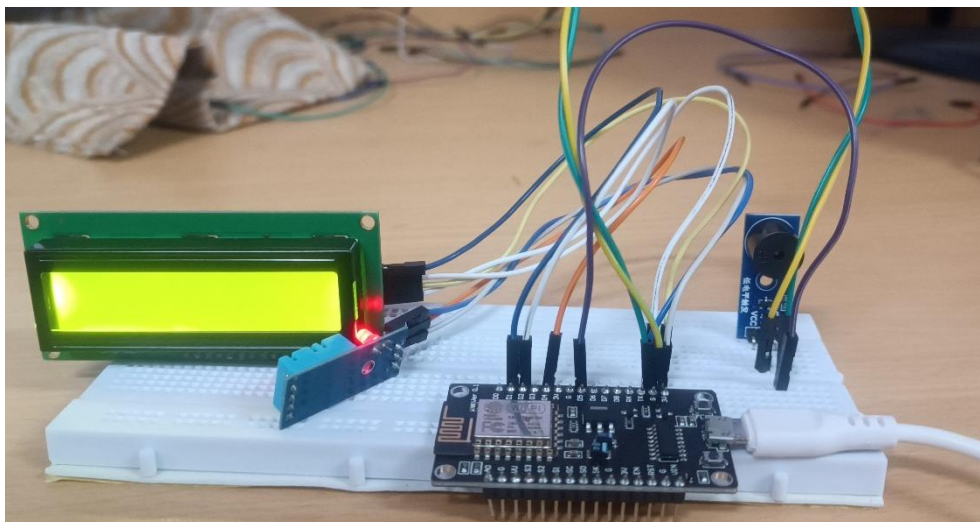
Prototype Implementation:

- The prototype was implemented using an ESP8266 NodeMCU, DHT11 temperature and humidity sensor, OLED display, green LED, red LED and buzzer. The DHT11 sensor was placed inside the refrigerator model to continuously measure the internal temperature and humidity.
- The sensor readings are collected and processed by the ESP8266. The measured temperature and humidity values are displayed on the OLED display for real-time monitoring. The ESP8266 compares the readings with predefined threshold values to determine the refrigerator's operating condition.
- Under normal conditions, the green LED is activated. If the temperature or humidity exceeds the predefined limit, the red LED and buzzer are activated to provide an immediate warning.
- The ESP8266 can also use its built-in Wi-Fi connectivity to transmit the sensor data to an IoT/cloud platform for remote monitoring. The prototype continuously updates the readings and alerts the user whenever abnormal conditions are detected.

The prototype follows the data flow:

DHT11 Sensor → ESP8266 → Data Processing → OLED Display + LED/Buzzer Alert → Wi-Fi → Cloud Monitoring

The implemented prototype demonstrates a **low-cost, real-time and automated solution** for monitoring refrigerator conditions and improving food preservation.



Results and Performance:

- The developed Smart Refrigerator Monitoring System successfully monitored the internal temperature and humidity using the DHT11 sensor and ESP8266. The sensor readings were processed in real time and displayed on the OLED display, allowing the user to continuously observe the refrigerator conditions.
- The system successfully identified normal and abnormal conditions based on predefined temperature and humidity thresholds. During normal conditions, the green LED indicated that the refrigerator environment was suitable for food preservation. When the measured values exceeded the defined limits, the red LED and buzzer provided an immediate warning.

Performance Analysis

Parameter	Result
Temperature Monitoring	Successfully measured
Humidity Monitoring	Successfully measured
Local Display	Real-time OLED display
Normal Condition	Green LED activated
Abnormal Condition	Red LED + buzzer activated
Response	Continuous and real-time
Wireless Connectivity	ESP8266 Wi-Fi supported
Remote Monitoring	IoT/Cloud integration supported

Overall, the prototype demonstrated a low-cost, reliable and real-time monitoring solution for refrigerator conditions. The system can help detect abnormal temperature and humidity conditions early, thereby supporting better food preservation and reducing food spoilage. Future improvements can include more accurate sensors, door-open detection, compressor monitoring and automated energy-efficiency analysis.

